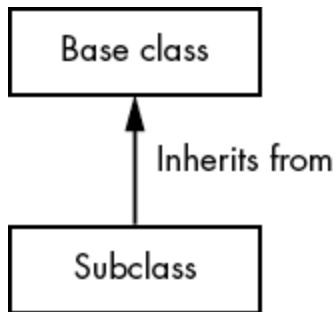


- *Base class* and *derived class*
- *Parent class* and *child class*

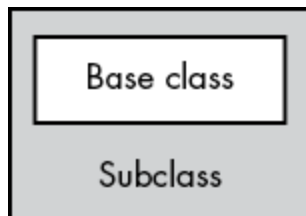
[Figure 10-1](#) is a standard diagram that shows this relationship.



[Figure 10-1](#): A subclass inherits from a base class.

A subclass inherits all of the methods and instance variables defined in a base class.

[Figure 10-2](#) provides a different, perhaps more accurate, way to think of the relationship between the two classes.



[Figure 10-2](#): A base class is incorporated into a subclass.

As the implementer, you can think of the base class as being incorporated into the subclass. That is, the base class actually becomes part of the larger subclass. As a client of a subclass, you think about the subclass as a single unit and do not need to know that the base class is there at all.

When discussing inheritance, we often say there is an *is a* relationship between a subclass and a base class. For example, a student is a person, an orange is a fruit, a car is a vehicle, and so on. The subclass is a specialized version of the base class that inherits all the properties and behavior of the base class, but also provides additional details and functionality.